Problem 4.29: A Beginner's Walkthrough

Isospin Cross-Section Ratios for $\pi N \to K\Sigma$

Built from the ground up — assumes rusty angular momentum

The Problem

Find the ratio of cross sections for:

(a) $\pi^- + p \to K^0 + \Sigma^0$

(b) $\pi^- + p \to K^+ + \Sigma^-$

(c) $\pi^+ + p \to K^+ + \Sigma^+$

First when $I = 3/2$ dominates, then when $I = 1/2$ dominates.

What are we even doing here?

These three reactions look different (different charges flying around), but they're all the *same* underlying strong-force process: a pion hits a nucleon and produces a kaon and a sigma. The strong force doesn't care about electric charge—it only cares about **isospin**. So the cross sections for all three reactions are *related*, and we can find their ratios using nothing but algebra. No complicated dynamics needed.

Part 0: Angular Momentum Refresher

Before touching isospin, let's dust off the angular momentum rules you'll need. Everything in this problem is a direct copy of spin/angular momentum algebra—just with different labels.

Refresher: The Key Rules

Rule 1: Quantum numbers j and m.
Any angular momentum is described by two numbers:

- j = the *total* angular momentum (always ≥ 0; can be 0, 1/2, 1, 3/2, ...)
- m = the z-component (ranges from $-j$ to $+j$ in integer steps)

So if $j = 1$, then $m \in \{-1, 0, +1\}$ — that's $2j + 1 = 3$ states.
If $j = 1/2$, then $m \in \{-1/2, +1/2\}$ — that's 2 states.
If $j = 0$, then $m = 0$ only — just 1 state.

Rule 2: The triangle rule for combining two angular momenta.
When you combine j_1 and j_2, the total J can be:

$$J = |j_1 - j_2|, \quad |j_1 - j_2| + 1, \quad \ldots, \quad j_1 + j_2 \tag{1}$$

Every integer (or half-integer) step in that range is allowed.

Rule 3: m values are additive.
The total z-component is always $M = m_1 + m_2$. No exceptions.

Rule 4: Clebsch–Gordan (CG) coefficients.
To express a product state $|m_1, m_2\rangle$ in terms of total-J states $|J, M\rangle$, you need CG coefficients:

$$|m_1, m_2\rangle = \sum_J \underbrace{\langle J, M \,|\, j_1, m_1;\, j_2, m_2\rangle}_{\text{CG coefficient}} |J, M\rangle \tag{2}$$

These are just numbers you can look up in a table or derive with the lowering operator method.

Quick example to check you're with me

Combine spin-1/2 with spin-1/2:

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0 \tag{3}$$

The triangle rule gives $J = |1/2 - 1/2|$ to $1/2 + 1/2$, i.e., $J = 0$ or 1.
State count: $2 \times 2 = 4$ product states, and $(2 \cdot 1 + 1) + (2 \cdot 0 + 1) = 3 + 1 = 4$. ✓

The state $|m_1 = +1/2,\ m_2 = +1/2\rangle$ has $M = +1$. Only $J = 1$ can have $M = +1$ (since $J = 0$ requires $M = 0$). So this state is *pure* $J = 1$. This is the **"stretched state"** trick—we'll use it repeatedly.

Part 1: What is Isospin?

Isospin is a copy of angular momentum algebra, applied to particle physics. The strong force treats the u and d quarks as two states of the same object, just like spin-up and spin-down:

$$u \longleftrightarrow \text{"isospin up"}: \quad I = \frac{1}{2}, \quad I_3 = +\frac{1}{2} \tag{4}$$

$$d \longleftrightarrow \text{"isospin down"}: \quad I = \frac{1}{2}, \quad I_3 = -\frac{1}{2} \tag{5}$$

Important conventions

- The strange quark s has $I = 0$. It's invisible to isospin.
- For **antiquarks**, I_3 flips sign: $\bar{u}$ has $I_3 = -1/2$, and $\bar{d}$ has $I_3 = +1/2$.
- I_3 is always additive: $I_3(\text{hadron}) = \sum I_3(\text{quarks})$.
- Total I follows from the triangle rule, just like spin.

Why isospin is useful

The strong interaction conserves isospin:

$$[H_{\text{strong}},\ \mathbf{I}] = 0 \tag{6}$$

This means the scattering amplitude T can only connect states with the *same* total I. Different charge channels of the same reaction share the same few amplitudes $\mathcal{A}_I$, weighted by different CG coefficients. That's how we'll relate the three cross sections.

Part 2: Assigning Isospin to Every Particle

Let's go through each particle in the problem, one by one.

The Pion: π^+, π^0, π^- — Isospin $I = 1$

Take π^+, which is made of $u\bar{d}$:

- u has $I = 1/2$, $\bar{d}$ has $I = 1/2$
- Triangle rule: $I = 0$ or 1
- $I_3 = (+1/2) + (+1/2) = +1$
- $I = 0$ can only have $I_3 = 0$, so $I_3 = +1$ **doesn't fit**
- Therefore π^+ **must** have $I = 1$ (stretched state argument!)

The other pions are partners in the same $I = 1$ triplet:

Pion	Quarks	I	I_3
π^+	$u\bar{d}$	1	$+1$
π^0	$(u\bar{u} - d\bar{d})/\sqrt{2}$	1	0
π^-	$d\bar{u}$	1	-1

The Nucleons: p, n — Isospin $I = 1/2$

The proton is uud. Three quarks, each with $I = 1/2$:

- Triangle rule (applied twice): possible totals are $I = 3/2$ or $I = 1/2$
- $I_3(p) = +1/2 + 1/2 - 1/2 = +1/2$
- Both $I = 3/2$ and $I = 1/2$ can have $I_3 = +1/2$, so the stretched state trick **doesn't help here**
- We need physics: the proton and neutron form a doublet (2 particles with similar mass), so $2I + 1 = 2$, giving $I = 1/2$

- The $I = 3/2$ slot is taken by the Δ baryons ($\Delta^{++}, \Delta^{+}, \Delta^{0}, \Delta^{-}$)

Nucleon	**Quarks**	$\boldsymbol{I}$	$\boldsymbol{I_3}$
p	uud	$1/2$	$+1/2$
n	udd	$1/2$	$-1/2$

The Kaons: K^+, K^0 — Isospin $I = 1/2$

K^+ is $u\bar{s}$. The $\bar{s}$ has $I = 0$, so only the u contributes:

- Only one light quark $\Rightarrow$ $I = 1/2$ automatically
- $I_3 = +1/2 + 0 = +1/2$

Kaon	**Quarks**	$\boldsymbol{I}$	$\boldsymbol{I_3}$
K^+	$u\bar{s}$	$1/2$	$+1/2$
K^0	$d\bar{s}$	$1/2$	$-1/2$

The Sigmas: Σ^+, Σ^0, Σ^- — Isospin $I = 1$

Σ^+ is uus. The s has $I = 0$, so only u and u contribute:

- Two u quarks, each $I = 1/2$. Triangle rule: $I = 0$ or 1
- $I_3 = +1/2 + 1/2 = +1$
- Only $I = 1$ fits $\Rightarrow$ stretched state! Σ^+ has $I = 1$

The other sigmas are in the same triplet. (The Λ^0, also uds, takes the $I = 0$ slot.)

Sigma	**Quarks**	$\boldsymbol{I}$	$\boldsymbol{I_3}$
Σ^+	uus	1	$+1$
Σ^0	uds	1	0
Σ^-	dds	1	-1

Summary: what we need going forward

Particle	I	I_3
$\pi^{\pm}$	1	± 1
p	1/2	$+1/2$
K^+	1/2	$+1/2$
K^0	1/2	$-1/2$
Σ^+	1	$+1$
Σ^0	1	0
Σ^-	1	-1

In practice, you'd look these up in the PDG. The derivations above are so you understand *why* the values are what they are.

Part 3: What Isospin Channels Exist?

Now we combine the particles in the initial and final states.

Initial state: $\pi + N$ (pion + nucleon)

Pion has $I = 1$, nucleon has $I = 1/2$. Triangle rule:

$$I_{\text{total}} = |1 - \tfrac{1}{2}| \text{ to } 1 + \tfrac{1}{2} = \frac{1}{2} \text{ to } \frac{3}{2} \tag{7}$$

So the possible total isospins of the πN system are:

$$\boxed{I = \frac{1}{2} \quad \text{or} \quad I = \frac{3}{2}} \tag{8}$$

Final state: $K + \Sigma$ (kaon + sigma)

Kaon has $I = 1/2$, sigma has $I = 1$. Triangle rule:

$$I_{\text{total}} = |1/2 - 1| \text{ to } 1/2 + 1 = \frac{1}{2} \text{ to } \frac{3}{2} \tag{9}$$

Same two possibilities: $I = 1/2$ or $3/2$.

This is not a coincidence!

Both the initial and final states have the same allowed isospins (1/2 and 3/2). This *must* be the case—the strong interaction conserves I, so the initial and final states must be able to "talk to each other" in the same isospin channels. If the final state could only have $I = 3/2$ but the initial state also had $I = 5/2$, that extra channel would simply be irrelevant.

The key physics

Since the strong force conserves total isospin, the scattering amplitude has **two independent pieces**:

- $\mathcal{A}_{3/2}$ = the amplitude for scattering in the $I = 3/2$ channel
- $\mathcal{A}_{1/2}$ = the amplitude for scattering in the $I = 1/2$ channel

These are complicated numbers that encode all the difficult strong-force dynamics. But we *never need to compute them*. We just need to know how much each reaction "overlaps" with each channel—and that's given by CG coefficients.

Part 4: The CG Coefficients We Need

We need to decompose the physical particle states into states of definite total isospin. This is where the CG coefficients come in.

Building the CG table for $1 \otimes 1/2$ from scratch

I'll derive everything using the lowering operator, step by step.

Refresher: The lowering operator

The lowering operator J_- decreases m by 1:

$$J_-|j,m\rangle = \sqrt{j(j+1) - m(m-1)}\,|j,m-1\rangle \tag{10}$$

For a two-particle system: $J_- = J_{1-} + J_{2-}$ (it acts on each particle independently).

Step A: Start at the top.

The maximum I_3 from $1 \otimes 1/2$ is $I_3 = 1 + 1/2 = 3/2$. Only $I = 3/2$ accommodates this, so:

$$\left|\frac{3}{2}, +\frac{3}{2}\right\rangle = |1,+1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \tag{11}$$

Step B: Lower to $I_3 = +1/2$.

Apply $I_- = I_{1-} + I_{2-}$ to both sides of (11).

Left side:

$$I_- \left|\frac{3}{2}, +\frac{3}{2}\right\rangle = \sqrt{\frac{3}{2}\cdot\frac{5}{2} - \frac{3}{2}\cdot\frac{1}{2}}\left|\frac{3}{2}, +\frac{1}{2}\right\rangle = \sqrt{3}\left|\frac{3}{2}, +\frac{1}{2}\right\rangle \tag{12}$$

Right side:

I_{1-} acts on $|1,+1\rangle$: $\sqrt{1\cdot 2 - 1\cdot 0}\,|1,0\rangle = \sqrt{2}\,|1,0\rangle$

I_{2-} acts on $|1/2,+1/2\rangle$: $\sqrt{1/2\cdot 3/2 - 1/2\cdot(-1/2)}\,|1/2,-1/2\rangle = 1\cdot|1/2,-1/2\rangle$

So:

$$\sqrt{3}\left|\frac{3}{2}, +\frac{1}{2}\right\rangle = \sqrt{2}\,|1,0\rangle\left|\frac{1}{2}, +\frac{1}{2}\right\rangle + |1,+1\rangle\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{13}$$

Divide by $\sqrt{3}$:

$$\boxed{\left|\frac{3}{2},+\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|1,0\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\,|1,+1\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle} \tag{14}$$

Step C: Find $|1/2,+1/2\rangle$ by orthogonality.

In the $I_3 = +1/2$ subspace, there are two product states:

$$|1,0\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle \quad \text{and} \quad |1,+1\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{15}$$

We've used one linear combination for $|3/2,+1/2\rangle$. The $|1/2,+1/2\rangle$ state must be *orthogonal*:

$$\boxed{\left|\frac{1}{2},+\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\,|1,0\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\,|1,+1\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle} \tag{16}$$

(Swap the coefficients and flip one sign. The Condon–Shortley convention fixes which sign flips.)

Step D: Lower to $I_3 = -1/2$.

Apply I_- to (14). After the same kind of algebra (which I'll spare you), the results are:

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\,|1,-1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\,|1,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{17}$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = -\sqrt{\frac{2}{3}}\,|1,-1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\,|1,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{18}$$

Are these right? Let's check!

Each equation should be normalized (squared coefficients add to 1):

- (17): $1/3 + 2/3 = 1$ ✓
- (18): $2/3 + 1/3 = 1$ ✓

And they should be orthogonal (dot product = 0):

$$\left(-\sqrt{\frac{2}{3}}\right)\left(\sqrt{\frac{1}{3}}\right) + \left(\sqrt{\frac{1}{3}}\right)\left(\sqrt{\frac{2}{3}}\right) = -\frac{\sqrt{2}}{3} + \frac{\sqrt{2}}{3} = 0 \quad \checkmark \tag{19}$$

Part 5: Decomposing the Initial States

Now we use the CG table to write the physical particle states in terms of total isospin.

$\pi^- p$ initial state (reactions a and b)

In isospin language:

$$|\pi^-\rangle = |1,-1\rangle, \qquad |p\rangle = \left|\frac{1}{2},+\frac{1}{2}\right\rangle \tag{20}$$

So $|\pi^- p\rangle = |1,-1\rangle|1/2,+1/2\rangle$, which has $I_3 = -1 + 1/2 = -1/2$.

We need to express this product state in terms of $|I, I_3\rangle$. We do this by **inverting** equations (17) and (18). Those equations express total-isospin states in terms of product states. We want the reverse.

Equations (17) and (18) form a system of two equations with two unknowns. Solving for $|1,-1\rangle|1/2,+1/2\rangle$:

Multiply (17) by $\sqrt{1/3}$ and (18) by $-\sqrt{2/3}$ and add:

$$\begin{aligned}
&\sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \\
&= \sqrt{\frac{1}{3}}\left(\sqrt{\frac{1}{3}}\,|1,-1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\,|1,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle\right) \\
&\quad - \sqrt{\frac{2}{3}}\left(-\sqrt{\frac{2}{3}}\,|1,-1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\,|1,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle\right) \\
&= \left(\frac{1}{3}+\frac{2}{3}\right)|1,-1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle + \left(\frac{\sqrt{2}}{3}-\frac{\sqrt{2}}{3}\right)|1,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle \\
&= |1,-1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle
\end{aligned} \tag{21}$$

So:

$$\boxed{|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle} \tag{22}$$

What this means physically

The $\pi^- p$ state is a **superposition** of $I = 3/2$ and $I = 1/2$. It's "1/3 in the $I = 3/2$ channel and 2/3 in the $I = 1/2$ channel" (in terms of probabilities, i.e., squared coefficients). This is why $\pi^- p$ reactions involve *both* amplitudes $\mathcal{A}_{3/2}$ and $\mathcal{A}_{1/2}$.

$\pi^+ p$ initial state (reaction c)

$$|\pi^+ p\rangle = |1,+1\rangle\left|\frac{1}{2},+\frac{1}{2}\right\rangle \tag{23}$$

This has $I_3 = +1 + 1/2 = +3/2$. Now ask: which total isospin can have $I_3 = +3/2$?

- $I = 1/2$? Maximum $I_3 = +1/2$. **Nope, too small.**
- $I = 3/2$? Maximum $I_3 = +3/2$. **Yes!**

So $\pi^+ p$ is a **pure** $I = 3/2$ state:

$$\boxed{|\pi^+ p\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \tag{24}$$

This is the most important observation in the problem!

Because $\pi^+ p$ is *pure* $I = 3/2$, reaction (c) has **zero access** to the $I = 1/2$ amplitude. This means:

- When $I = 3/2$ dominates: reaction (c) is at full strength
- When $I = 1/2$ dominates: reaction (c) is **completely forbidden** ($\sigma_c = 0$)

Part 6: Decomposing the Final States

Now we do the same for the outgoing $K\Sigma$ states. There's one subtlety: the kaon has $I = 1/2$ and the sigma has $I = 1$, so we're coupling in the order $1/2 \otimes 1$ instead of $1 \otimes 1/2$.

Swapping the coupling order

The CG coefficients for $j_2 \otimes j_1$ are related to those for $j_1 \otimes j_2$ by:

$$\langle j_2, m_2;\, j_1, m_1 | J, M\rangle = (-1)^{j_1 + j_2 - J} \langle j_1, m_1;\, j_2, m_2 | J, M\rangle \tag{25}$$

For $j_1 = 1$, $j_2 = 1/2$: the phase is $(-1)^{3/2-J}$, which is $+1$ for $J = 3/2$ and -1 for $J = 1/2$.
Translation: the $I = 3/2$ coefficients stay the same, but the $I = 1/2$ coefficients all flip sign.

Using this, we can write down the final-state decompositions. I'll use the notation $|I_{3,K}\rangle|I_{3,\Sigma}\rangle$ for the product states.

$K^0\Sigma^0$ (final state of reaction a)

$$|K^0\Sigma^0\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle |1, 0\rangle \qquad I_3 = -\frac{1}{2} \tag{26}$$

Applying the coupling-order relation to the CG table we built:

$$\boxed{|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{27}$$

(The $I = 3/2$ coefficient keeps its sign from the $1 \otimes 1/2$ table; the $I = 1/2$ coefficient *flips*, turning $-$ into $+$.)

$K^+\Sigma^-$ (final state of reaction b)

$$|K^+\Sigma^-\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle |1,-1\rangle \qquad I_3 = -\frac{1}{2} \tag{28}$$

$$\boxed{|K^+\Sigma^-\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{29}$$

$K^+\Sigma^+$ (final state of reaction c)

$$|K^+\Sigma^+\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle |1,+1\rangle \qquad I_3 = +\frac{3}{2} \tag{30}$$

Same stretched-state argument as before: $I_3 = +3/2$ only fits in $I = 3/2$.

$$\boxed{|K^+\Sigma^+\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \tag{31}$$

Part 7: Computing the Amplitudes

Now we have all the ingredients. The amplitude for each reaction is:

$$\mathcal{M} = \langle \text{final}| \, T \, |\text{initial}\rangle = \sum_I \langle \text{final}|I, I_3\rangle \; \mathcal{A}_I \; \langle I, I_3|\text{initial}\rangle \tag{32}$$

In words: project the initial state onto each isospin channel, multiply by the amplitude for that channel, then project onto the final state. Let me do each reaction very explicitly.

Reaction (a): $\pi^- p \to K^0\Sigma^0$

Initial state (22): $|\pi^- p\rangle = \sqrt{\frac{1}{3}}|3/2\rangle - \sqrt{\frac{2}{3}}|1/2\rangle$

(I'm abbreviating $|I, I_3\rangle$ as $|I\rangle$ since $I_3 = -1/2$ throughout.)

Final state (27): $|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}|3/2\rangle + \sqrt{\frac{1}{3}}|1/2\rangle$

The amplitude:

The T operator only connects same-I states. So:

$$\begin{aligned}
\mathcal{M}_a &= \langle K^0\Sigma^0| \, T \, |\pi^- p\rangle \\
&= \underbrace{\sqrt{\frac{2}{3}}}_{\text{final, } I=3/2} \times \underbrace{\mathcal{A}_{3/2}}_{\text{amplitude}} \times \underbrace{\sqrt{\frac{1}{3}}}_{\text{initial, } I=3/2} + \underbrace{\sqrt{\frac{1}{3}}}_{\text{final, } I=1/2} \times \underbrace{\mathcal{A}_{1/2}}_{\text{amplitude}} \times \underbrace{\left(-\sqrt{\frac{2}{3}}\right)}_{\text{initial, } I=1/2} \\
&= \frac{\sqrt{2}}{3}\,\mathcal{A}_{3/2} \; - \; \frac{\sqrt{2}}{3}\,\mathcal{A}_{1/2}
\end{aligned} \tag{33}$$

$$\boxed{\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} - \mathcal{A}_{1/2}\right)} \tag{34}$$

Sign convention note

Griffiths writes $\mathcal{M}_a = \frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$. The difference is just a sign convention in the $K\Sigma$ CG coefficients (the coupling-order phase). It doesn't affect the final cross-section ratios, because only $|\mathcal{M}|^2$ matters, and we're taking limits where one of $\mathcal{A}_{3/2}$ or $\mathcal{A}_{1/2}$ vanishes. I'll adopt Griffiths' convention from here on.

Adopting Griffiths' convention:

$$\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right) \tag{a}$$

Reaction (b): $\pi^- p \to K^+\Sigma^-$

Same initial state. Final state (29): $|K^+\Sigma^-\rangle = \sqrt{1/3}|3/2\rangle - \sqrt{2/3}|1/2\rangle$

$$\begin{aligned}\mathcal{M}_b &= \sqrt{\frac{1}{3}} \times \mathcal{A}_{3/2} \times \sqrt{\frac{1}{3}} + \left(-\sqrt{\frac{2}{3}}\right) \times \mathcal{A}_{1/2} \times \left(-\sqrt{\frac{2}{3}}\right) \\ &= \frac{1}{3}\,\mathcal{A}_{3/2} + \frac{2}{3}\,\mathcal{A}_{1/2}\end{aligned} \tag{35}$$

In Griffiths' convention:

$$\boxed{\mathcal{M}_b = \frac{1}{3}\left(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right)} \tag{b}$$

Reaction (c): $\pi^+ p \to K^+\Sigma^+$

Initial state (24): pure $|3/2, +3/2\rangle$, coefficient $= 1$.
Final state (31): pure $|3/2, +3/2\rangle$, coefficient $= 1$.

$$\boxed{\mathcal{M}_c = 1 \times \mathcal{A}_{3/2} \times 1 = \mathcal{A}_{3/2}} \tag{c}$$

Reaction (c) is the simplest!

Both initial and final states are pure $I = 3/2$, so the amplitude is just $\mathcal{A}_{3/2}$ with no mixing. No CG coefficients to worry about (they're both 1). This is the beauty of the stretched state.

Part 8: Taking Cross-Section Ratios

The cross section is proportional to $|\mathcal{M}|^2$:

$$\sigma_a \propto |\mathcal{M}_a|^2 = \frac{2}{9}\left|\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right|^2 \tag{36}$$

$$\sigma_b \propto |\mathcal{M}_b|^2 = \frac{1}{9}\left|\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right|^2 \tag{37}$$

$$\sigma_c \propto |\mathcal{M}_c|^2 = \left|\mathcal{A}_{3/2}\right|^2 \tag{38}$$

Case 1: $I = 3/2$ dominates ($\mathcal{A}_{1/2} \approx 0$)

Set $\mathcal{A}_{1/2} = 0$:

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{3/2}|^2 \tag{39}$$

$$\sigma_b \propto \frac{1}{9}|\mathcal{A}_{3/2}|^2 \tag{40}$$

$$\sigma_c \propto |\mathcal{A}_{3/2}|^2 \tag{41}$$

The $|\mathcal{A}_{3/2}|^2$ cancels in the ratio:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = \frac{2}{9} : \frac{1}{9} : 1 = 2 : 1 : 9} \tag{42}$$

Case 2: $I = 1/2$ dominates ($\mathcal{A}_{3/2} \approx 0$)

Set $\mathcal{A}_{3/2} = 0$:

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{1/2}|^2 \tag{43}$$

$$\sigma_b \propto \frac{1}{9} \cdot 4|\mathcal{A}_{1/2}|^2 = \frac{4}{9}|\mathcal{A}_{1/2}|^2 \tag{44}$$

$$\sigma_c \propto 0 \tag{45}$$

$$\boxed{\sigma_a : \sigma_b : \sigma_c = \frac{2}{9} : \frac{4}{9} : 0 = 1 : 2 : 0} \tag{46}$$

The $\sigma_c = 0$ prediction

This is the most dramatic result: **the reaction $\pi^+ p \to K^+\Sigma^+$ is completely forbidden when $I = 1/2$ dominates.**

Why? Because $\pi^+ p$ is pure $I = 3/2$. If the $I = 3/2$ amplitude is turned off, the $\pi^+ p$ system simply has nothing to scatter through. It's like trying to fit a square peg in a round hole—the quantum numbers don't match.

This is an experimentally testable prediction. At energies dominated by N^* resonances (which have $I = 1/2$), you should see $\sigma(\pi^+ p \to K^+\Sigma^+) \approx 0$.

Part 9: Summary and Sanity Checks

All results in one place

	Amplitude	$I = 3/2$ **dom.**	$I = 1/2$ **dom.**
(a) $\pi^- p \to K^0 \Sigma^0$	$\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$	2	1
(b) $\pi^- p \to K^+ \Sigma^-$	$\frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2})$	1	2
(c) $\pi^+ p \to K^+ \Sigma^+$	$\mathcal{A}_{3/2}$	9	0

Consistency check: amplitude relation

The three amplitudes satisfy a model-independent identity:

$$\mathcal{M}_c = \sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b \tag{47}$$

Proof:

$$\begin{aligned}\sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b &= \sqrt{2}\cdot\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}) + \frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}) \\ &= \frac{2}{3}\mathcal{A}_{3/2} + \frac{2}{3}\mathcal{A}_{1/2} + \frac{1}{3}\mathcal{A}_{3/2} - \frac{2}{3}\mathcal{A}_{1/2} \\ &= \mathcal{A}_{3/2} = \mathcal{M}_c \quad \checkmark\end{aligned} \tag{48}$$

This relation holds for *any* values of $\mathcal{A}_{3/2}$ and $\mathcal{A}_{1/2}$—it's a consequence of isospin symmetry alone.

Quick ratio check for $I = 3/2$ dominance

If $\mathcal{A}_{1/2} = 0$: $\mathcal{M}_a = \frac{\sqrt{2}}{3}\mathcal{A}_{3/2}$, $\mathcal{M}_b = \frac{1}{3}\mathcal{A}_{3/2}$, $\mathcal{M}_c = \mathcal{A}_{3/2}$.
$|\mathcal{M}_a|^2 : |\mathcal{M}_b|^2 : |\mathcal{M}_c|^2 = \frac{2}{9} : \frac{1}{9} : 1$. Multiply by 9: $2 : 1 : 9$. ✓

Quick ratio check for $I = 1/2$ dominance

If $\mathcal{A}_{3/2} = 0$: $\mathcal{M}_a = \frac{\sqrt{2}}{3}\mathcal{A}_{1/2}$, $\mathcal{M}_b = -\frac{2}{3}\mathcal{A}_{1/2}$, $\mathcal{M}_c = 0$.
$|\mathcal{M}_a|^2 : |\mathcal{M}_b|^2 : |\mathcal{M}_c|^2 = \frac{2}{9} : \frac{4}{9} : 0$. Multiply by 9/2: $1 : 2 : 0$. ✓

The recipe for any isospin problem like this

1. **Look up** (or derive) the isospin assignments: I and I_3 for every particle. (PDG is your friend.)
2. **Triangle rule**: what total isospins are possible for the initial and final states?
3. **CG decomposition**: express each physical state as a sum over $|I, I_3\rangle$ states.

(Look up the CG table or derive with the lowering operator.)

4. **Amplitude**: $\mathcal{M} = \sum_I (\text{final CG coeff.}) \times \mathcal{A}_I \times (\text{initial CG coeff.})$
5. **Ratio**: $\sigma \propto |\mathcal{M}|^2$. In limiting cases, one $\mathcal{A}_I$ vanishes, giving pure-number ratios.